handedness randomly when observed, the gloves work in tandem, even if widely separated: if one chooses left, the other chooses right, and vice versa.

5. Quantum mechanics makes predictions about the microworld that agree fantastically well with experimental observations. On this, there is universal agreement. Nevertheless, because the detailed features of quantum mechanics, as discussed in this chapter, differ significantly from those of common experience, and, relatedly, as there are different mathematical formulations of the theory (and different formulations of how the theory spans the gap between the microworld of phenomena and the macroworld of measured results), there isn't consensus on how to *interpret* various features of the theory (and various puzzling data which the theory, nevertheless, is able to explain mathematically), including issues of nonlocality. In this chapter, I have taken a particular point of view, the one I find most convincing based on current theoretical understanding and experimental results. But, I stress here that not everyone agrees with this view, and in a later endnote, after explaining this perspective more fully, I will briefly note some of the other perspectives and indicate where you can read more about them. Let me also stress, as we will discuss later, that the experiments contradict Einstein's belief that the data could be explained solely on the basis of particles always possessing definite, albeit hidden, properties *without any use or mention of nonlocal entanglement.* However, the failure of this perspective only rules out a local universe. It does not rule out the possibility that particles have such definite hidden features.

6. For the mathematically inclined reader, let me note one potentially misleading aspect of this description. For multiparticle systems, the probability wave (the wavefunction, in standard terminology) has essentially the same interpretation as just described, but is defined as a function on the *configuration space* of the particles (for a single particle, the configuration space is isomorphic to real space, but for an N-particle system it has 3N dimensions). This is important to bear in mind when thinking about the question of whether the wavefunction is a real physical entity or merely a mathematical device, since if one takes the former position, one would need to embrace the reality of configuration space as well—an interesting variation on the themes of Chapters 2 and 3. In relativistic quantum field theory, the fields can be defined in the